DIAGRAM 1: SYSTEM ARCHITECTURE DIAGRAM

Dual-Stage Hybrid Machine Learning Engine with SHAP & Spatial GIS Integration

LAYER 1: PRESENTATION TIER (React 19 + Tailwind CSS + Recharts + Leaflet GIS)

Valuation UI
Single & Batch Mode
- Property Form
- CSV Bulk Uploader

SHAP Waterfall
Explainable AI Card
- Positive Premiums
- Negative Penalties

Spatial GIS Map
Leaflet Interactive
- Centroid Scatter
- Radius to CBD Bands

Admin & Benchmark
Dynamic Control
- CSV Retraining Fit
- Metric Benchmarks

HTTP Dispatch / REST State

LAYER 2: APPLICATION SERVICES & DATA INGESTION PIPELINE

API Latency Dispatcher
Simulates 0-2000ms latency

CSV Ingestion Engine
Auto schema mapping & parse

Runtime State Cache
Reactive localStorage store

jsPDF Report Generator
Audit-ready certificate engine

Feature Vector (X_structural, X_spatial)

LAYER 3: CORE ML ENGINE — DUAL-STAGE HYBRID INFERENCE & EXPLAINABILITY SUBSYSTEM

Stage 1: Hedonic Baseline (OLS)
*P_linear = b0 + Sum(bi*Xi) + Area*Rate*
- Locality base rates (Rs/sq.ft)
- Macroeconomic baseline bounds
- High interpretability (R2 = 0.742)
- Linear parameter stability

Residuals

Stage 2: Residual Learner (XGBoost)
R_pred = XGBoost(Spatial, Structural)
- Fits error residuals (Actual - P_lin)
- Non-linear cross-interactions
- Distance decay to CBD & amenities
- Gradient boosted decision trees

Hybrid Synthesis Ensemble
P_final = P_linear + R_pred
- Achieves R2 = 0.932 benchmark
- Reduces MAE to Rs 8.45 Lakhs
- Calculates 95% Confidence Band
- Eliminates black-box opacity

Explainable AI Subsystem (Shapley Additive Explanations - SHAP)
P_final = E[P] + Sum(Phi_j) --> Exact Feature-by-Feature Monetary Decomposition (in Rs Lakhs)
- Positive Contributor: +Rs 14.20 L for Indiranagar Centroid | +Rs 6.50 L for 3 BHK Spatial Configuration
- Negative Deductions: -Rs 4.80 L for Age > 12 yrs | -Rs 2.10 L for Distance from Metro > 8 km
- Delivers visual waterfall data to frontend & institutional PDF certificate generator

Read / Write Persistent Artifacts

LAYER 4: GEOSPATIAL & HISTORICAL TRANSACTION DATA STORE

Property Transactions
Bengaluru Housing Registry
- Area, BHK, Age, Price
- Custom uploaded CSVs

Neighborhood Centroids
GIS Coordinate Database
- Lat, Long coordinates
- CBD Euclidean Distance (km)

Model Weights & Metrics
Dynamic Weight Store
- OLS slopes & XGBoost tree
- 5-fold CV error cache

Key Innovation Note for College Evaluation:

This 4-tier architecture eliminates the classic black-box trade-off. Stage 1 guarantees baseline financial bounds (Hedonic OLS), Stage 2 fits spatial residuals via XGBoost, and the

DIAGRAM 2: EXECUTION WORKFLOW & DATA PIPELINE

Step-by-Step Runtime Data Processing Pipeline from User Inputs to SHAP & Map Delivery

PHASE 1: CLIENT INGESTION & PARAMETER CAPTURE

Option A: Single Property Mode

Direct User Parameter Entry

- Inputs: Area (Sq.Ft.), BHK, Bathrooms, Age, Floors
- Locality Selector (Indiranagar, Koramangala, etc.)
- Parking, Furnishing status, Amenities checklist

Option B: Batch CSV Upload Mode

Institutional Bulk Appraisal

- Upload multi-row property transaction CSV
- Automated column detection & schema alignment
- Supports real-time bulk valuation up to 2,000 rows

PHASE 2: PREPROCESSING & GEOSPATIAL FEATURE ENGINEERING

2.1 Data Validation

Null check, numeric range bounds, ty

2.2 Spatial Lookup

Geocoding centroid & CBD distance:

2.3 Derived Features

Area-per-Bedroom density, Bath-to-B

2.4 Normalization

Min-Max scaling of numeric metrics & categorical

Engineered Matrix [X]

PHASE 3: DUAL-STAGE HYBRID INFERENCE & ENSEMBLE SYNTHESIS

Stage 3A: OLS Hedonic Baseline Calculation

*P_linear = Area * BaseRate_loc + Sum(w_i * X_i)*

- Establishes neighborhood baseline pricing floor
- Accounts for primary square footage & basic features

Stage 3B: XGBoost Residual Correction

Res_pred = XGBoost(Spatial, Amenities, Age)

- Predicts non-linear variance and neighborhood premium
- Corrects floor-level, luxury, and proximity deviations

P_linear **Res_pred**

Stage 3C: Hybrid Synthesis & Valuation Band

Final Valuation = P_linear + Res_pred | Confidence Band = [P - 6.8%, P + 6.8%]

- Produces robust property valuation with R2 = 0.932 and MAE = Rs 8.45 Lakhs

Synthesized Valuation

PHASE 4: SHAP EXPLAINABILITY & MONETARY DECOMPOSITION

Local & Global Shapley Value Calculation: Phi_i = Sum [|S|!(|F|-|S|-1)! / |F|!] * [f(S U {i}) - f(S)]

Decomposes every rupee of price change into human-readable positive premiums and negative deductions

- Positive Value Adds (+): Location Primacy (Indiranagar: +Rs 14.2 L), Built-up Area (+Rs 42.5 L), 2 Covered Parking (+Rs 4.0 L)
- Negative Deductions (-): Property Age > 15 yrs (-Rs 5.2 L), High Distance to Transit Metro Station (-Rs 2.8 L)
- Provides plain-language audit narrative for home buyers, real estate brokers, and loan underwriters

Output Payloads

PHASE 5: MULTI-CHANNEL INTERACTIVE USER PRESENTATION

Valuation Card

Instant Price Display

- Predicted Price (Lakhs)
- Range [Min, Max]

SHAP Waterfall

Visual Breakdown

- Green / Red bars
- Interactive hover

Leaflet GIS Map

Spatial Context

- Centroid map marker
- CBD distance rings

Audit PDF Report

Downloadable File

- jsPDF vector file
- Official Certificate

DIAGRAM 3 & 4: USE CASE MODEL & DATABASE SCHEMA

Actor Interactions, Functional Scope, and Relational Entity Architecture

DIAGRAM 3: USE CASE MODEL & ACTOR ROLE INTERACTIONS

Actor: User / Buyer

Standard Role Access

- Register & Sign In
- Submit Property Parameters
- Explore GIS Gravity Map
- Download Valuation PDF

Standard User Use Cases (Valuation & Explainability)

[UC1] Register & Authenticate User Session
[UC2] Single Property Valuation Input
[UC3] Batch CSV Bulk Property Valuation
[UC4] Render Local SHAP Waterfall Breakdown
[UC5] Inspect Spatial Clusters & CBD Distance
[UC6] Compare Multi-Model Performance Benchmarks

Actor: Administrator

Elevated System Control

- All User Privileges
- Upload Custom Property CSV
- Trigger In-Browser Retrain
- Configure API Latency

Administrator Use Cases (Machine Learning Operations & Tuning)

[UC7] Ingest Custom Property CSV Dataset
[UC8] Execute Dynamic Client-Side Model Retraining
[UC9] Compute 5-Fold Cross-Validation Metrics (R2, RMSE)
[UC10] Adjust Simulation Network Latency (0ms - 2000ms)

DIAGRAM 4: ENTITY-RELATIONSHIP (ER) DATA SCHEMA

USER

- *** id (PK, Int)**
- name (String)
- email (String, Unique)
- password (Hashed)
- role (Admin/Member)
- token (Session)

PROPERTY

- *** id (PK, Int)**
- area (Sq.Ft.)
- bedrooms, baths, age
- location (FK)
- distanceToCBD (km)
- price (Target, Lakhs)

NEIGHBORHOOD_CONFIG

- *** location (PK, String)**
- baseRate (Rs/Sq.Ft.)
- latitude (Float)
- longitude (Float)
- distanceFromCenter (km)

PREDICTION_RESULT

- *** prediction_id (PK, Int)**
- user_id (FK)
- final_price (Lakhs)
- lower_bound, upper_bound
- explanation (Text)

Key Schema Relationships:

• USER (1) ----< (N) PREDICTION_RESULT | • NEIGHBORHOOD_CONFIG (1) ----< (N) PROPERTY | • PREDICTION (1) ----< (N) SHAP_VALUES

PROJECT EXPERIMENTAL BENCHMARK SUMMARY TABLE

Model Architecture	Mean Absolute Error (MAE)	Root Mean Sq. Error (RMSE)	R^2 Score
Ordinary Least Squares (OLS Hedonic)	Rs 18.42 Lakhs	Rs 24.18 Lakhs	0.742
Standalone Random Forest Regressor	Rs 12.65 Lakhs	Rs 17.84 Lakhs	0.851
Standard Gradient Boosting (GBM)	Rs 11.20 Lakhs	Rs 15.92 Lakhs	0.876
Proposed Hybrid (OLS + XGBoost + SHAP)	**Rs 8.45 Lakhs**	**Rs 12.10 Lakhs**	**0.932 (Best)**